

Day 01 – Unit 1 : Micro-Economic Analysis

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A. Demand Analysis

1. Marshallian Demand (Marshall, 1890)

- **Cardinal utility** — utility numbers (utils) mein measure hoti hai
- Equilibrium : $MU_x/P_x = MU_y/P_y$ (Law of Equi-Marginal Utility)
- Demand fn. $x = f(P, M) \rightarrow$ **Uncompensated demand**
- MU of money **constant** maani $\rightarrow$ income effect ignore
- **Consumer Surplus** ka concept Marshall ki den; Giffen exception possible

2. Hicksian Demand (Hicks & Allen, 1934)

- **Ordinal utility** — IC + Budget Line; equilibrium par $MRS_{xy} = P_x/P_y$
- **Compensated demand** $h(P, U)$ — real income / utility constant
- Expenditure minimisation se derive $\rightarrow$ sirf **Substitution Effect**
- Curve **hamesha downward sloping** — koi Giffen exception nahi
- **Slutsky Equation** : Price Effect = Substitution Effect + Income Effect

★ Hicks vs Slutsky Compensation — exam favourite

Hicks	Slutsky
Income itni adjust ki consumer original IC par wapas aaye	Income itni adjust ki consumer original bundle khareed sake
Utility constant	Purchasing power constant
IC observe nahi hoti $\rightarrow$ non-observable	Observable & measurable $\rightarrow$ empirically superior

3. Revealed Preference Theory (Samuelson, 1938)

- **Behaviouristic** — na cardinal, na ordinal utility ki zarurat
- A aur B dono affordable the, A chuna $\rightarrow$ A "revealed preferred" over B
- **WARP** = consistency | **SARP** (Houthakker) = transitivity $\rightarrow$ IC derive ho jati hai
- **"Fundamental Theorem of Consumption Theory"** : Substitution effect hamesha negative
- **Limitation** : indifference handle nahi karta, sirf strict preference

4. Lancaster's Theory (1966)

- Utility **goods se nahi, goods ki characteristics** se milti hai
- Car $\rightarrow$ utility speed, comfort, mileage se (car se nahi)
- Tools : Characteristics space + **Efficiency Frontier**
- Product differentiation aur naye products explain karta hai

Base	Marshallian	Hicksian	Revealed Pref.
<i>Propounder</i>	Marshall (1890)	Hicks-Allen (1934)	Samuelson (1938)
<i>Utility</i>	Cardinal	Ordinal	Koi nahi
<i>Constant kya</i>	Money income	Real income / Utility	Kuch nahi
<i>Demand fn.</i>	$x(P,M)$ Uncompensated	$h(P,U)$ Compensated	Observed choice
<i>Derived from</i>	Utility maximisation	Expenditure minimisation	Choice observation
<i>Slope</i>	Giffen mein upward	Hamesha downward	Hamesha downward

♦ Yaad Rakho

- **Normal good** → Hicksian curve **steeper** (kam elastic)
- **Inferior good** → Hicksian curve **flatter**
- **Shephard's Lemma** → Hicksian demand | **Roy's Identity** → Marshallian demand

B. Theory of Production

Law of Variable Proportions (Short Run)

Stage	TP	MP	AP	Naam
I	Increasing rate	Badhta → peak	Badhta, end pe max	Increasing Returns
II	Diminishing rate	Ghatta, end pe $MP=0$	Ghatta (MP se zyada)	Rational Stage
III	Ghatta hai	Negative	Ghatta	Negative Returns

★ TP - AP - MP Golden Relations

- $MP > AP \rightarrow AP$ badhta | $MP < AP \rightarrow AP$ ghatta
- **$MP = AP \rightarrow AP$ MAXIMUM** (MP curve AP ko uske highest point par cut karta hai)
- **$MP = 0 \rightarrow TP$ MAXIMUM** | MP negative → TP girta hai
- Producer hamesha **Stage II** mein produce karta hai

Isoquant Analysis (Long Run)

- **$MRTS(LK) = MPL / MPK$** = isoquant ka slope; Iso-cost slope = w/r
- **Producer Equilibrium** : $MRTS = w/r \Leftrightarrow MPL/w = MPK/r$ (Least Cost Combination)
- **Ridge lines** = economic region ki boundary (jahan $MP = 0$)
- **Expansion Path** (scale line) = budget badhne par equilibrium points ka locus

σ (elasticity of substitution)	Isoquant shape	Function
$\sigma = 0$	L-shaped (right angle)	Leontief / Fixed proportion
$\sigma = 1$	Convex	Cobb-Douglas
$\sigma = \infty$	Straight line	Perfect substitutes
$\sigma = \text{constant}$	—	CES (Arrow-Chenery-Minhas-Solow, 1961)

Returns to Scale — Cobb-Douglas $Q = A \cdot L^a K^b$

- $a + b > 1 \rightarrow IRS$ | **$a + b = 1 \rightarrow CRS$** (linear homogeneous) | $a + b < 1 \rightarrow DRS$
- **Euler's Theorem / Product Exhaustion** : CRS ke under **$Q = L(MPL) + K(MPK)$**

C. Cost Curves

Short Run — $TC = TFC + TVC$

Curve	Shape	Key Point
TFC	Horizontal	Output se independent
TVC	Inverse-S	Origin se start
$AFC = TFC/Q$	Rectangular hyperbola	X-axis ko kabhi touch nahi karti
AVC, AC, MC	U-shaped	Law of Variable Proportions ki wajah se

★ Golden Cost Relations

- MC, AVC aur AC dono ko unke MINIMUM point par cut karta hai (niche se)
- $MC < AC \rightarrow AC$ ghatta | $MC > AC \rightarrow AC$ badhta | $MC = AC \rightarrow AC$ minimum
- AVC ka minimum, AC ke minimum se PEHLE aata hai
- $AC - AVC = AFC \rightarrow$ gap ghatta hai par kabhi zero nahi
- $MC = w/MPL$ aur $AVC = w/APL \rightarrow$ MP max par MC min, AP max par AVC min
- MC sirf TVC se derive $\rightarrow$ TFC ka MC par koi asar nahi

Long Run

- LAC = "Envelope Curve" / "Planning Curve" — SACs ko sirf tangent karti hai, cut nahi
- LAC ke minimum par : $LAC = SAC = LMC = SMC$ (optimum plant + optimum output)
- LAC ke falling part par $\rightarrow$ SAC apne min ke LEFT side touch (under-utilised)
- LAC ke rising part par $\rightarrow$ SAC apne min ke RIGHT side touch (over-utilised)
- LAC U-shaped kyun ? $\rightarrow$ Economies of Scale, phir Diseconomies of Scale
- Modern Theory (Bain, Andrews) : SAC = saucer-shaped, LAC = L-shaped, MES concept

D. Market Structures

Universal Rule : $MC = MR$ + MC curve MR ko niche se cut kare. Difference sirf AR/MR ke shape ka hai.

Base	Perfect Comp.	Monopoly	Monopolistic	Oligopoly
Sellers	Bahut zyada	Ek	Bahut	Few (2-10)
Product	Homogeneous	No substitute	Differentiated	Dono possible
Demand curve	Horizontal	Downward, less elastic	Downward, more elastic	Kinked / Indeterminate
AR — MR	$AR = MR = P$	$AR > MR$ (slope double)	$AR > MR$	MR discontinuous
LR profit	Normal only	Supernormal	Normal only	Supernormal possible
LR condition	$P = MC = AC(\min)$	$P > MC$	$P = AC$ par AC min par NAHI	Indeterminate
Capacity	Full / Optimum	Under-utilised	Excess Capacity	Excess capacity
Selling cost	Nil	Nominal	Maximum	Bahut zyada
Supply curve	MC above AVC min	Hota hi nahi	Nahi	Nahi

- Shut-down point = $P < AVC(\min)$ | Break-even = $P = AC(\min)$
- Amoroso-Robinson : $MR = P(1 - 1/e)$ | Lerner Index = $(P - MC)/P = 1/e$
- Monopolist hamesha elastic portion ($e > 1$) par produce karta hai
- Price Discrimination (Pigou) : 1st = perfect (poora surplus), 2nd = block, 3rd = market segmentation ($MR_1 = MR_2 = MC$; kam elastic market mein zyada price)

Oligopoly Models

Model	Assumption	Result
Cournot (1838)	Rival ka output constant	Har firm $1/3$, total $2/3$; n firms $\rightarrow n/(n+1)$
Bertrand (1883)	Rival ka price constant	$P = MC \rightarrow$ "Bertrand Paradox"
Edgeworth (1897)	Capacity constraint	Indeterminate, price oscillate
Stackelberg	Leader – Follower	Determinate; leader ko zyada profit
Chamberlin	Interdependence pehchani	Monopoly (joint profit max) solution
Sweezy (1939)	Kinked Demand	Price Rigidity

★ Kinked Demand (Sweezy) – most asked

- Kink ke upar wala part **elastic**, neeche wala **inelastic**
- Isse **MR curve mein discontinuity / gap** aata hai
- MC us gap ke andar shift kare $\rightarrow$ **price aur output UNCHANGED** = price stickiness
- **Stigler ki criticism** : price kyun nahi badalta ye batata hai, par price pehle set kaise hua ye nahi

- **Collusive** : Cartel (perfect / market-sharing) + Price Leadership — **dominant, low-cost, barometric**
- **Prisoner's Dilemma** $\rightarrow$ cheat karne ka incentive $\rightarrow$ cartel unstable
- **Nash Equilibrium** = koi firm akele deviate karke better off nahi ho sakti
- **Chamberlin (1933)** : product differentiation + **selling cost**; **dd** curve (flatter) & **DD** curve (steeper); "product group"
- **Excess Capacity Theorem** = Chamberlin, par "**Ideal Output**" concept = **R. F. Kahn**

E. General Equilibrium

- **Partial** = Marshall (ceteris paribus) | **General** = Walras, 1874 (simultaneous equations)
- **Walras' Law** : sabhi markets ki excess demand ka sum = 0 $\rightarrow$ agar **$n-1$** markets equilibrium mein hain to **n -th** apne aap equilibrium mein
- **Numeraire** $\rightarrow$ sirf **relative prices** determine hote hain, absolute nahi
- **Tâtonnement** = "groping"; actual trading sirf equilibrium price par $\rightarrow$ **no false trading**
- Existence ka formal proof $\rightarrow$ **Arrow & Debreu (1954)**, fixed-point theorem se

♦ $2 \times 2 \times 2$ Model – 3 Conditions

Condition	Formula	Tool
1. Efficiency in Exchange	$MRS^A = MRS^B$	Edgeworth Box $\rightarrow$ Contract Curve
2. Efficiency in Production	$MRTS^X = MRTS^Y$	Production Contract Curve $\rightarrow$ PPF
3. Overall / Product-mix	$MRS = MRT$	Top-level condition

- PPF slope = $MRT = MC_x / MC_y$, concave to origin
- GUPF + Social Welfare Function ka tangency = **Bliss Point**

F. Welfare Economics

Chain : Pigou → Pareto → Kaldor-Hicks-Scitovsky → Bergson-Samuelson → Arrow

- **Pigou (1920)** : cardinal utility; welfare ↑ jab national income ↑ aur distribution zyada equal ho; **Pigovian Tax** (negative externality) & **Pigovian Subsidy** (positive). **Robbins ki criticism** → interpersonal comparison unscientific
- **Pareto Optimality** : kisi ko better kiye bina koi worse nahi. Base = ordinal, value-judgement free.
Limitations : unique nahi (contract curve par infinite points) + distribution ke baare mein bilkul chup

★★ Two Fundamental Theorems

Theorem	Statement	Meaning
1st	Har perfectly competitive equilibrium Pareto optimal hota hai	"Invisible Hand" ka formal proof
2nd	Koi bhi Pareto optimal allocation achieve ki ja sakti hai — bas lump-sum redistribution of endowments	Efficiency aur Equity separable hain

Market Failure (conditions fail) : externalities, public goods, monopoly, asymmetric information

Criterion	Test
Kaldor (1939)	Gainners, losers ko compensate karke bhi better off rahen
Hicks (1940)	Losers, gainers ko bribe karke change rok nahi sakte
Scitovsky (1941)	Double Test — Kaldor test PASS aur reverse test FAIL (Scitovsky Paradox se bachne ke liye)
Little	Efficiency test + distribution bhi improve ho

★ Arrow's Impossibility Theorem (1951)

Individual preferences se consistent SWF banana **asambhav** hai jo ye 5 conditions saath satisfy kare :

1. Unrestricted domain 2. Pareto principle 3. Transitivity 4. **IIA** 5. Non-dictatorship

→ Sirf **Dictatorship** hi sab satisfy kar sakti hai. Related : **Condorcet Paradox** ($A > B$, $B > C$, par $C > A$)

- **Bergson-Samuelson SWF** : $W = f(U_1, U_2, \dots, U_n)$ — value judgement zaroori
- **Rawlsian SWF** = $\min(U_1, U_2, \dots)$ = **Maximin** | **Benthamite** = $\sum U_i$
- **Theory of Second Best (Lipsey & Lancaster, 1956)** : ek condition fail ho to baaki ko satisfy karna welfare badhaye — **ye zaroori nahi** → piecemeal reform khatarnak
- **Coase Theorem** : property rights clear + zero transaction cost → externality private bargaining se solve (Pigovian tax ki zarurat nahi)

G. Managerial Theories of the Firm

Background : Berle & Means (1932) — "Divorce of Ownership from Control" → **Principal-Agent Problem** → manager ka objective profit se alag.

Base	Baumol (1959)	Marris (1964)	Williamson (1963)
Maximise kya	Sales Revenue (TR)	Balanced Growth Rate	Managerial Utility
Constraint	Minimum profit	Job security / Valuation ratio	Minimum profit
Key condition	$MR = 0$ ($e = 1$)	$g_D = g_C$	$U = f(S, M, I_D)$
Signature term	Sales maximisation	Valuation ratio, Takeover raid	Expense Preference, Slack
Output / Profit	Output zyada, Price kam, Profit kam	Growth zyada, Profit kam	Staff zyada, Profit kam
Nature	Static, single period	Dynamic (growth)	Static

★ Baumol — Tax ka Effect (bahut pucha jata hai)

• **Lump-sum / Profit tax** → Profit maximiser : **price-output UNCHANGED**

→ Baumol firm : **output GHATTA, price BADHTA** (profit constraint tight)

• **Per unit tax** → dono mein output ↓, price ↑

• Sales maximiser, profit maximiser se **ZYADA advertisement** karta hai

Marris ke 3 Financial Ratios (mila kar = Financial Security Constraint) :

1. **Leverage/Debt ratio** — zyada nahi (insolvency) 2. **Liquidity ratio** — balance mein 3. **Retention ratio** — zyada retention = kam dividend = share price girega

Valuation Ratio (v) = Market value of shares / Book value of assets → **v kam = takeover raid ka khatra = naukri gayi**

Williamson ke Profit Concepts :

• Actual Profit = $R - C - S$ | **Reported Profit** = Actual - M (emoluments)

• **Discretionary Profit** = Actual - Minimum Profit - Tax

• S = Staff M = Managerial Emoluments/Slack I_D = Discretionary Investment

Related : H. A. Simon → "Satisficing" + Bounded Rationality | **Cyert & March** → firm = coalition, "Organisational Slack"

H. Theory of Limit Price

Limit Price = wo highest price jo existing firms charge kar sakti hain bina naye entrants ko attract kiye. Short-run profit sacrifice → long-run profit bachao.

Range : Competitive price < Limit Price < Profit-maximising price

• **Joe S. Bain (1949, 56)** — father of limit pricing + **4 Barriers to Entry** :

(i) Economies of Scale (ii) Absolute Cost Advantage (iii) Product Differentiation (iv) Capital Requirement

• **Sylos-Labini (1962)** — **Sylos Postulate** : entrant maanta hai ki purani firms apna **output constant** rakhengi

• **Modigliani (1958)** — Bain + Sylos-Labini ka formal model

Bain's Entry Type	Pricing
Easy entry	Limit pricing possible nahi
Ineffectively impeded	Entry hone do, profit-max price lo
Effectively impeded	LIMIT PRICE set karo
Blockaded entry	Monopoly price (limit price ki zarurat nahi)

Limit vs Predatory Pricing : Limit = entry rokna, price **ABOVE cost** | Predatory = maujooda rival ko bahar nikalna, price **BELOW cost**

Contestable Markets (Baumol-Panzar-Willig, 1982) : costless entry-exit → "hit & run" → competitive outcome bina zyada firms ke

Unit 1 – Naam Yaad Karo (One-Liners)

Exam se pehle raat ko sirf ye page dekh lena

Concept	Economist
Revealed Preference Theory	P. A. Samuelson
Characteristics Approach to demand	K. Lancaster
Compensated demand curve	J. R. Hicks
WARP ka extension — SARP	Houthakker
CES production function	Arrow-Chenery-Minhas-Solow
Product Exhaustion Theorem	Euler
Envelope Curve (LAC)	Jacob Viner
L-shaped LAC / Modern cost theory	Bain, Andrews
Excess Capacity + Selling Cost	E. H. Chamberlin
"Ideal Output" concept	R. F. Kahn
Kinked Demand Curve	Paul Sweezy
Full Cost / Mark-up Pricing	Hall & Hitch
Duopoly — output constant	Cournot
Duopoly — price constant	Bertrand
Leader-Follower duopoly	Stackelberg
Price Discrimination ke 3 degrees	A. C. Pigou
Degree of Monopoly Power index	Lerner
General Equilibrium	Leon Walras
Existence proof of GE	Arrow & Debreu (1954)
Old Welfare Economics / Externality tax	A. C. Pigou
Compensation — Double Test	Scitovsky
Social Welfare Function	Bergson & Samuelson
Impossibility Theorem	K. J. Arrow
Theory of Second Best	Lipsey & Lancaster
Property rights & externality	R. H. Coase
Sales Revenue Maximisation	W. J. Baumol
Growth Maximisation / Valuation Ratio	Robin Marris
Managerial Utility / Expense Preference	O. E. Williamson
Satisficing & Bounded Rationality	H. A. Simon
Behavioural Theory / Organisational Slack	Cyert & March
Barriers to Entry / Limit Price	Joe S. Bain
Sylos Postulate	Sylos-Labini
Divorce of Ownership & Control	Berle & Means (1932)
Contestable Market Theory	Baumol, Panzar & Willig

★ Top 5 Confusions — galati mat karna

1. Hicksian = expenditure minimisation, Marshallian = utility maximisation
2. Excess Capacity = **Chamberlin**, par Ideal Output = **Kahn**
3. Lump-sum tax : profit-max mein output unchanged, Baumol mein output ghatta
4. Limit price > competitive price, par < monopoly/profit-max price
5. Walras' Law ≠ Say's Law